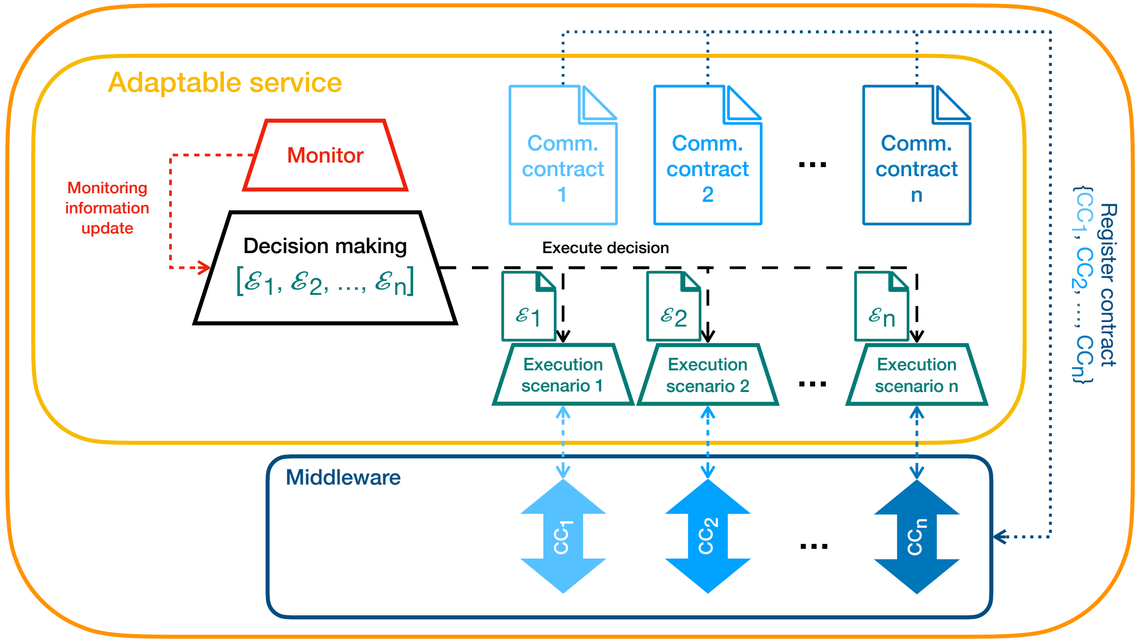
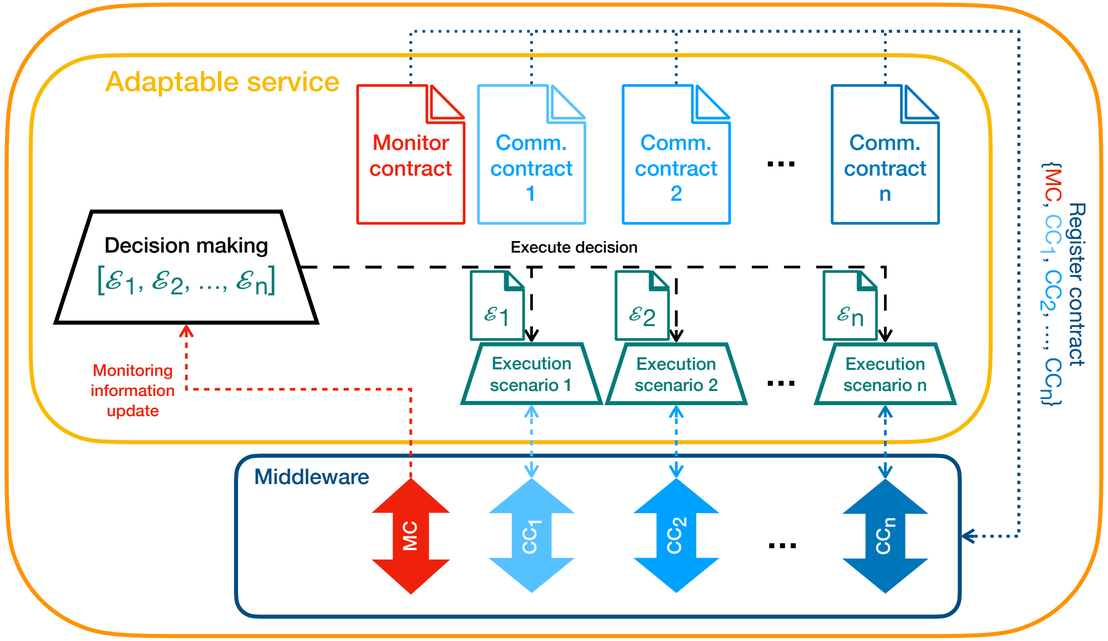
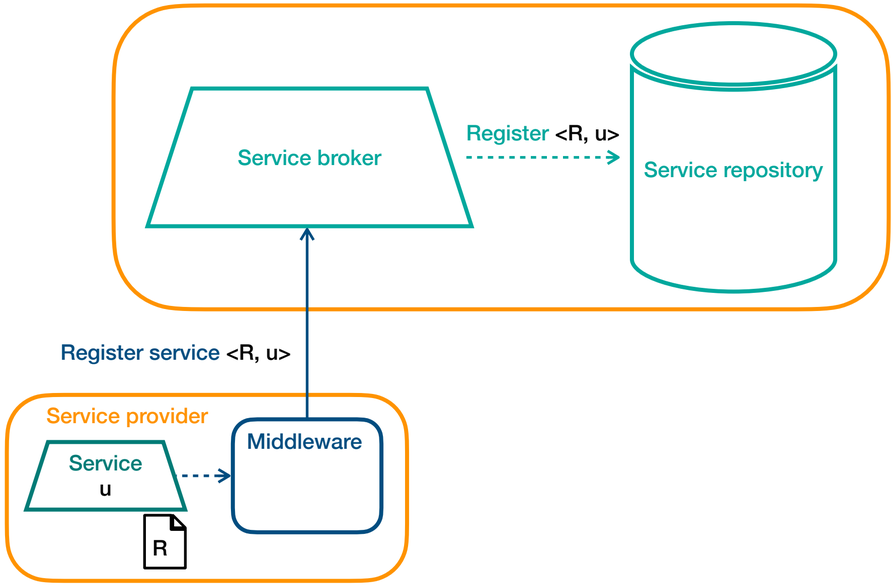
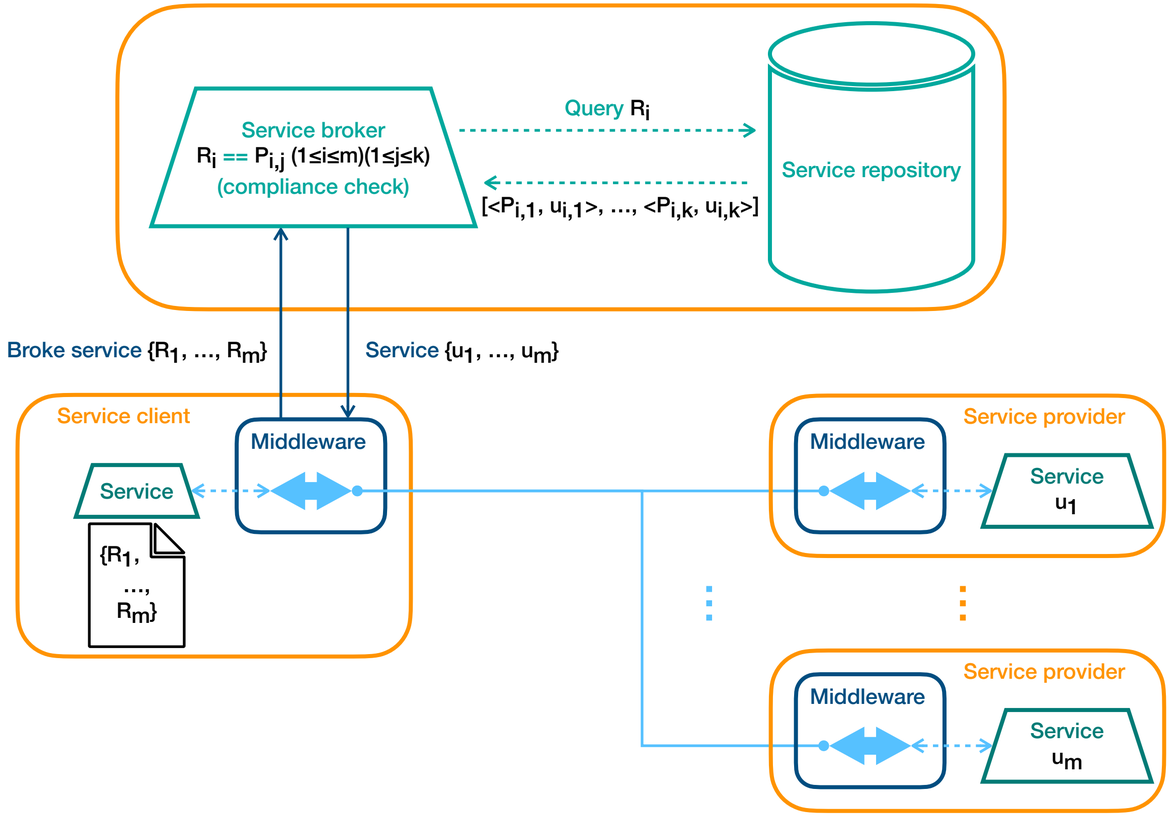
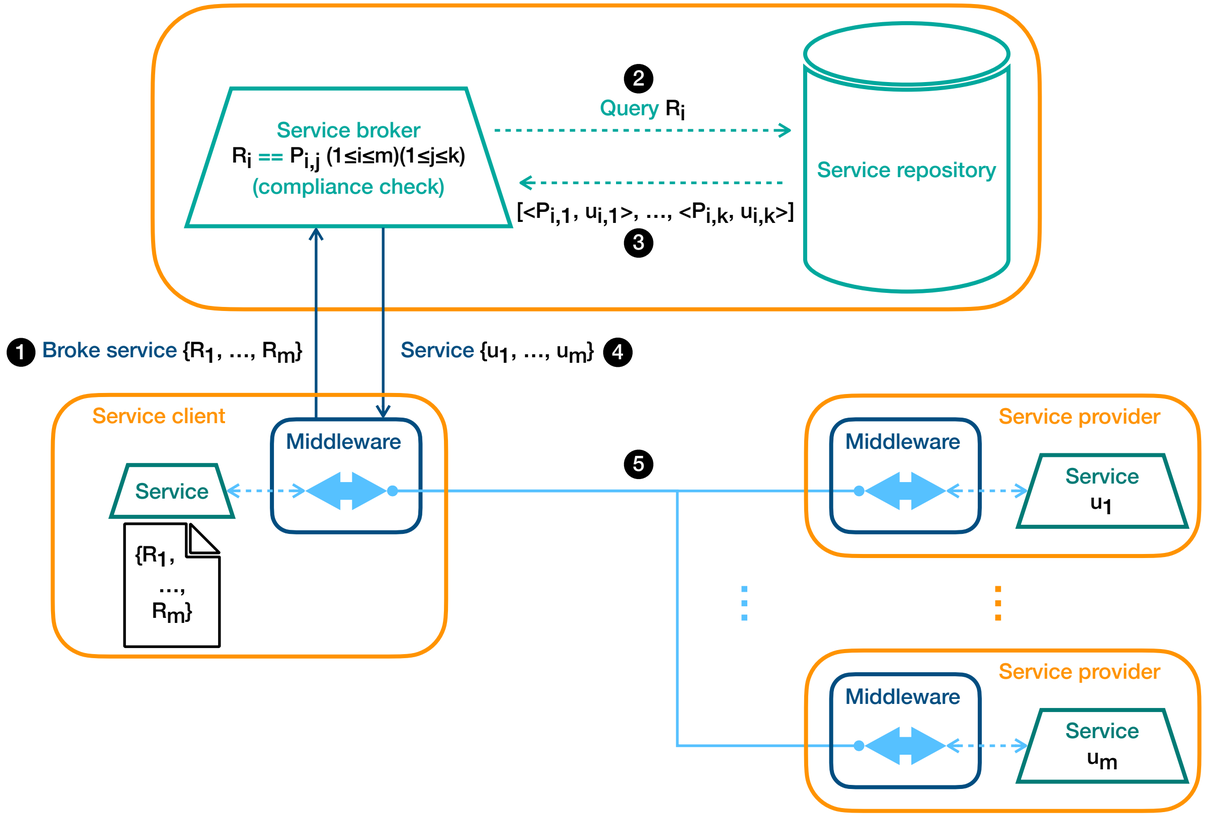
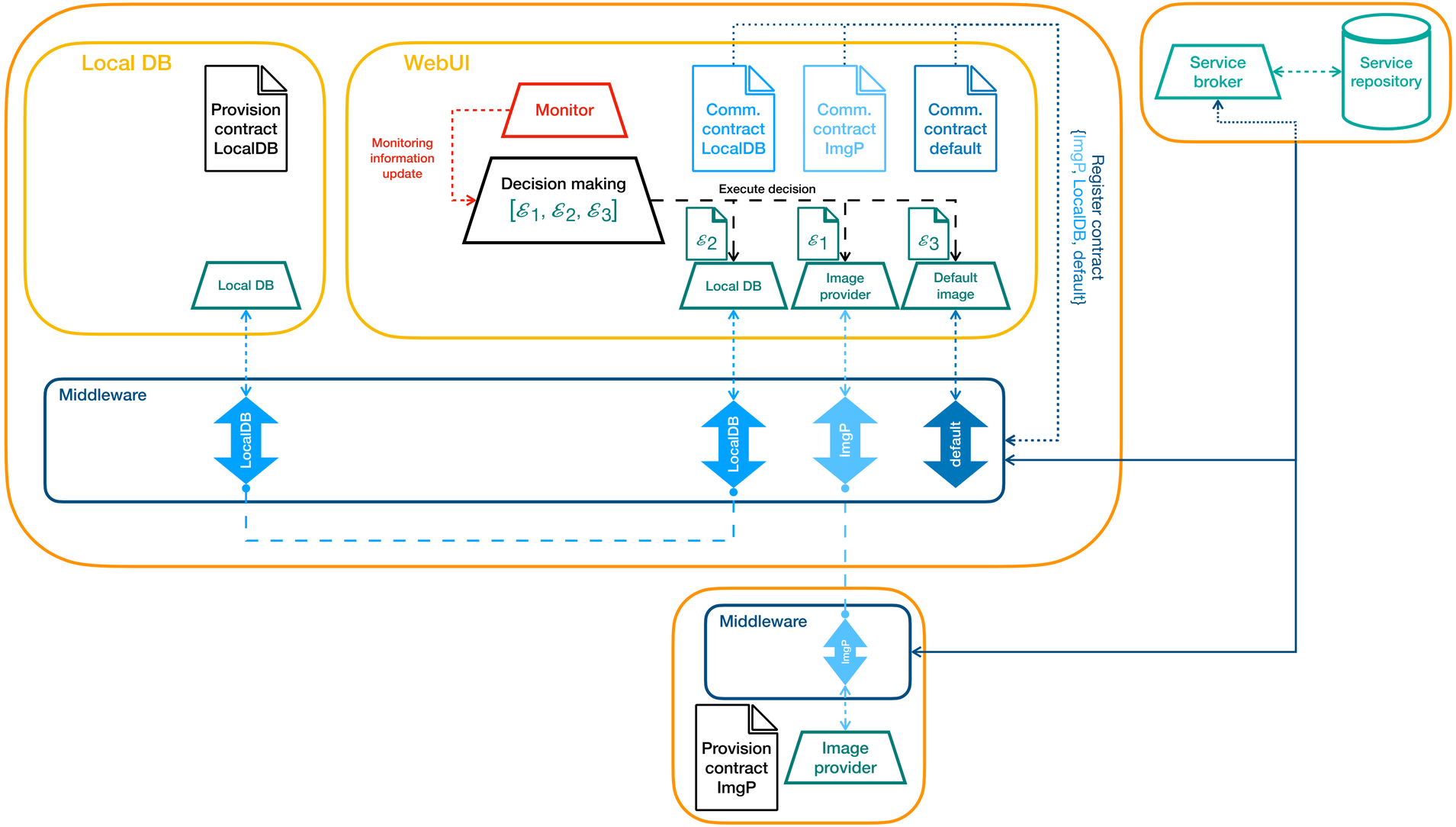

Adaptable service
Monitor
Monitoring information update
Decision making
[ℰ1, ℰ2, ..., ℰn]
Comm. contract 1
Comm. contract 2
...
Comm. contract n
Execute decision
ℰ1
ℰ2
ℰn
Execution scenario 1
Execution scenario 2
...
Execution scenario n
Middleware
CC1
CC2
...
CCn
Register contract
{CC1, CC2, ..., CCn}

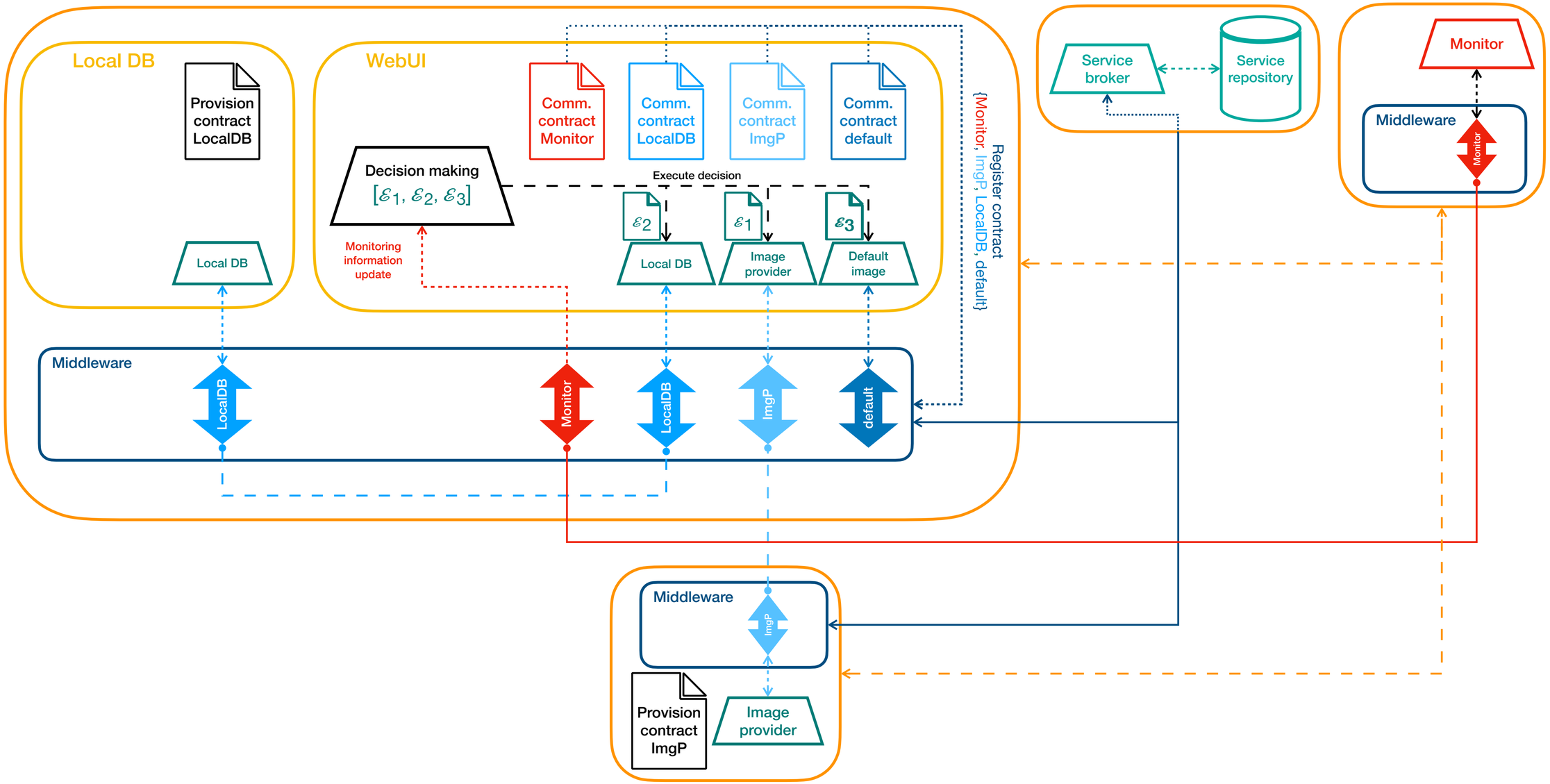

Adaptable service
Monitor contract
Comm. contract 1
Comm. contract 2
...
Comm. contract n
Decision making
[$\mathcal{E}_1$, $\mathcal{E}_2$, ..., $\mathcal{E}_n$]
Execute decision
$\mathcal{E}_1$
$\mathcal{E}_2$
$\mathcal{E}_n$
Execution scenario 1
Execution scenario 2
...
Execution scenario n
Monitoring information update
Middleware
MC
CC_1
CC_2
...
CC_n
Register contract
{MC, CC_1, CC_2, ..., CC_n}

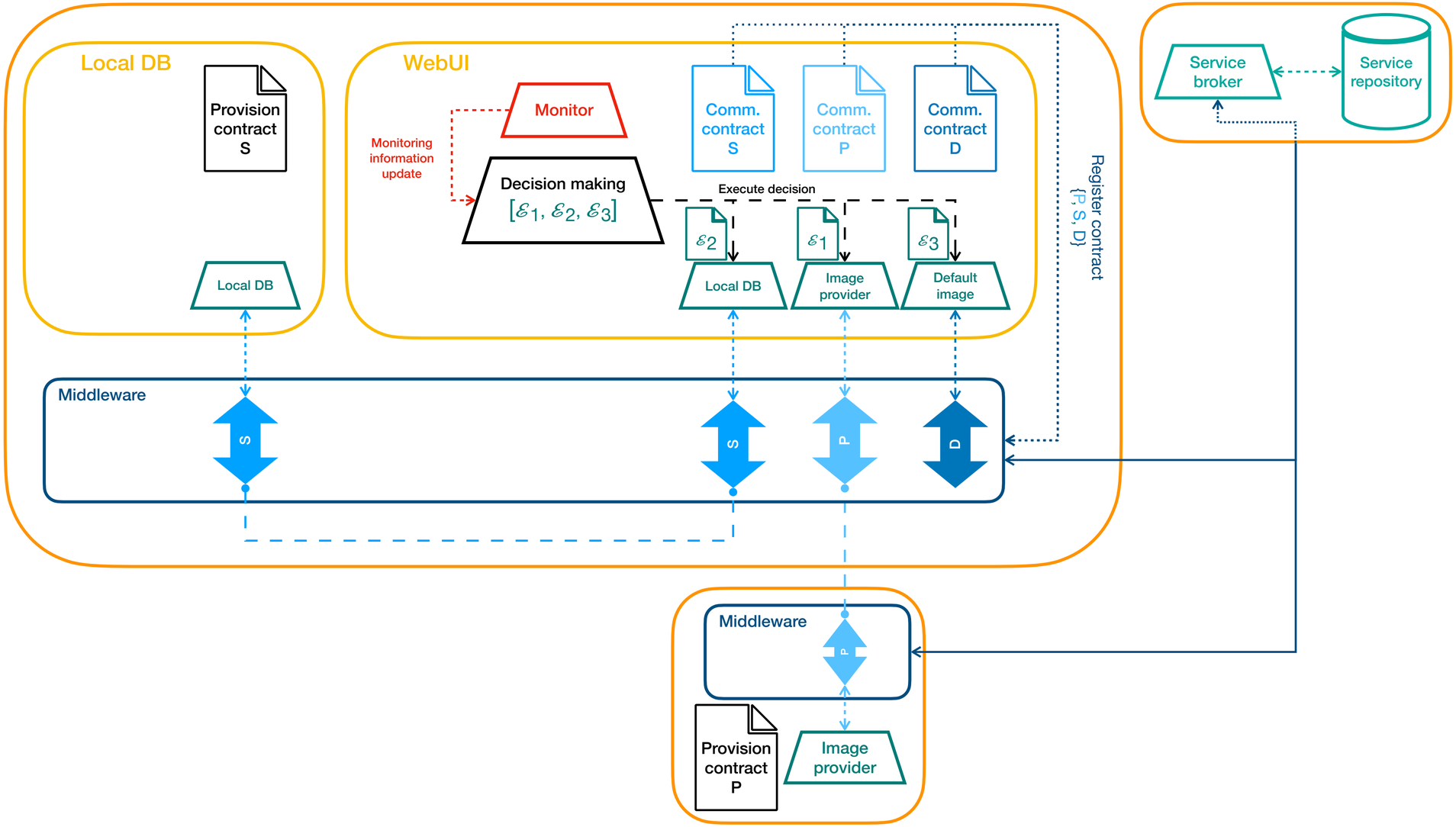

Service broker
Register <R, u>
Service repository
Register service <R, u>
Service provider
Middleware
Service
u
R

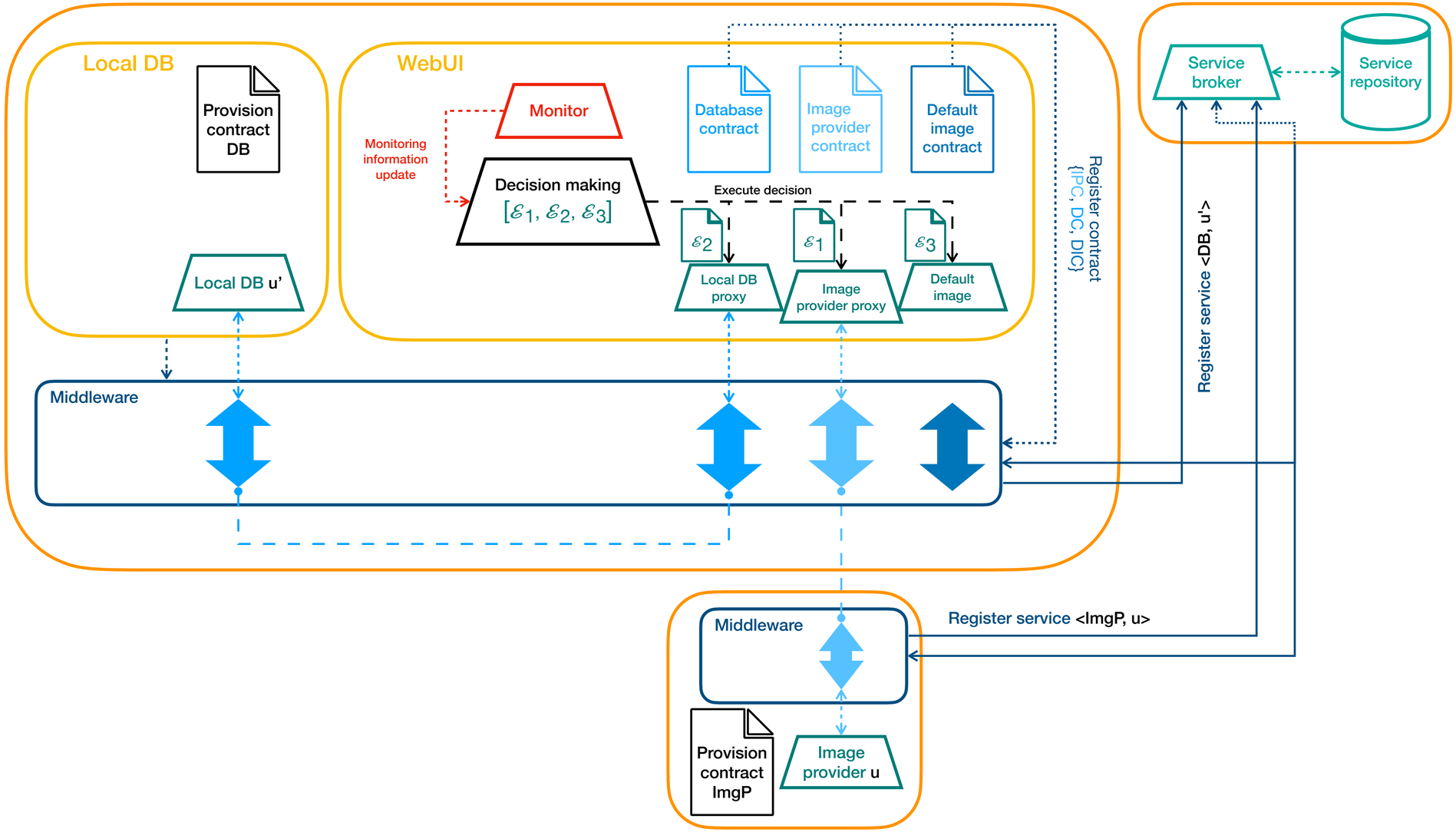

Service broker
$R_i == P_{i,j}$ $(1 \le i \le m)(1 \le j \le k)$
(compliance check)
Query R_i
$[<P_{i,1}, u_{i,1}>, \ldots, <P_{i,k}, u_{i,k}>]$
Service repository
Broke service $\{R_1, \ldots, R_m\}$
Service $\{u_1, \ldots, u_m\}$
Service client
Middleware
Service
$\{R_1, \ldots, R_m\}$
Service provider
Middleware
Service
u_1
Service provider
Middleware
Service
u_m

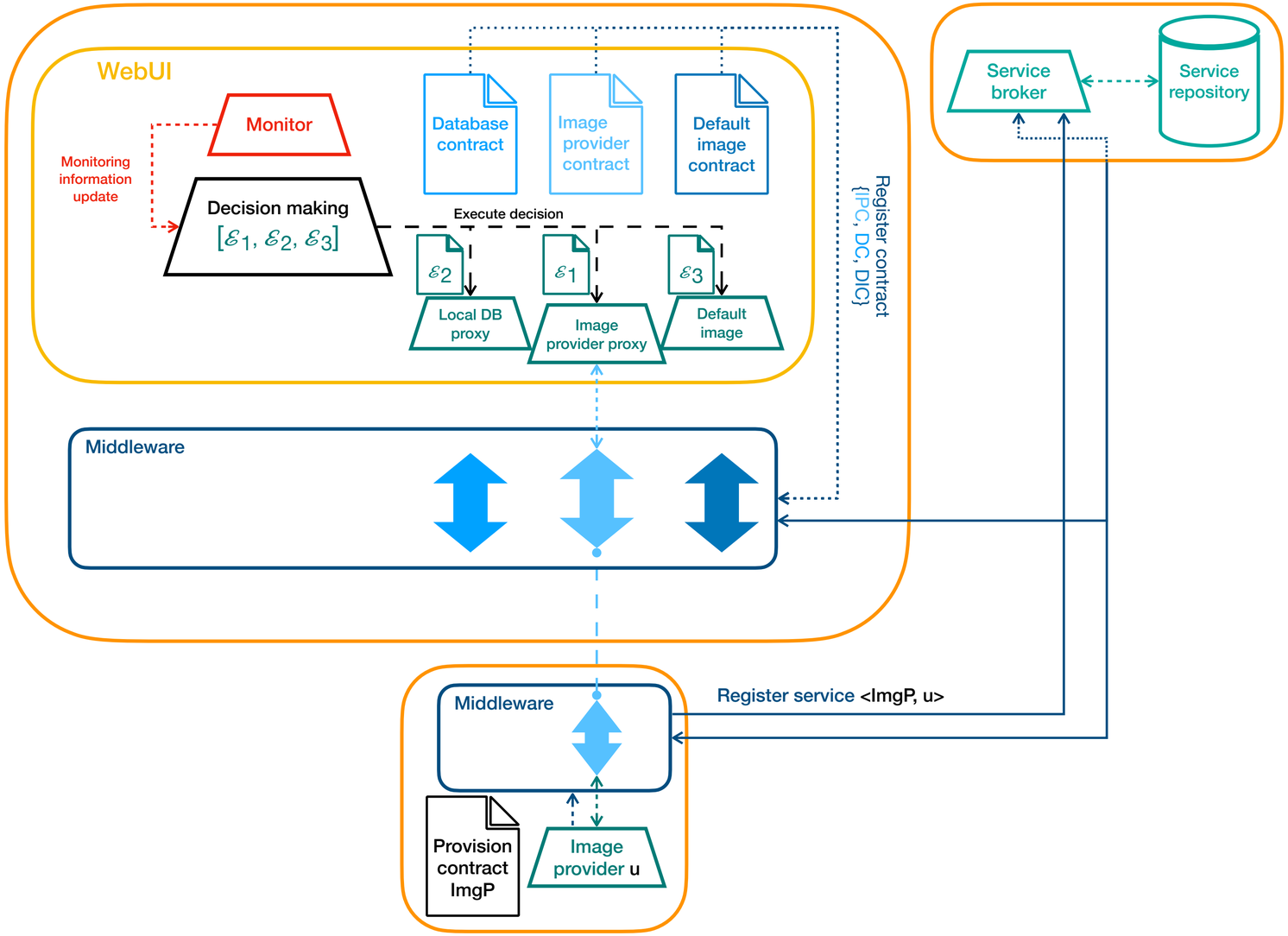

Service broker
$R_i == P_{i,j}$ $(1 \le i \le m)(1 \le j \le k)$
(compliance check)
2
Query R_i
Service repository
$[<P_{i,1}, u_{i,1}>, \ldots, <P_{i,k}, u_{i,k}>]$
3
1 Broke service $\{R_1, \ldots, R_m\}$
Service $\{u_1, \ldots, u_m\}$ 4
Service client
Middleware
Service
$\{R_1, \ldots, R_m\}$
5
Service provider
Middleware
Service
u_1
Service provider
Middleware
Service
u_m

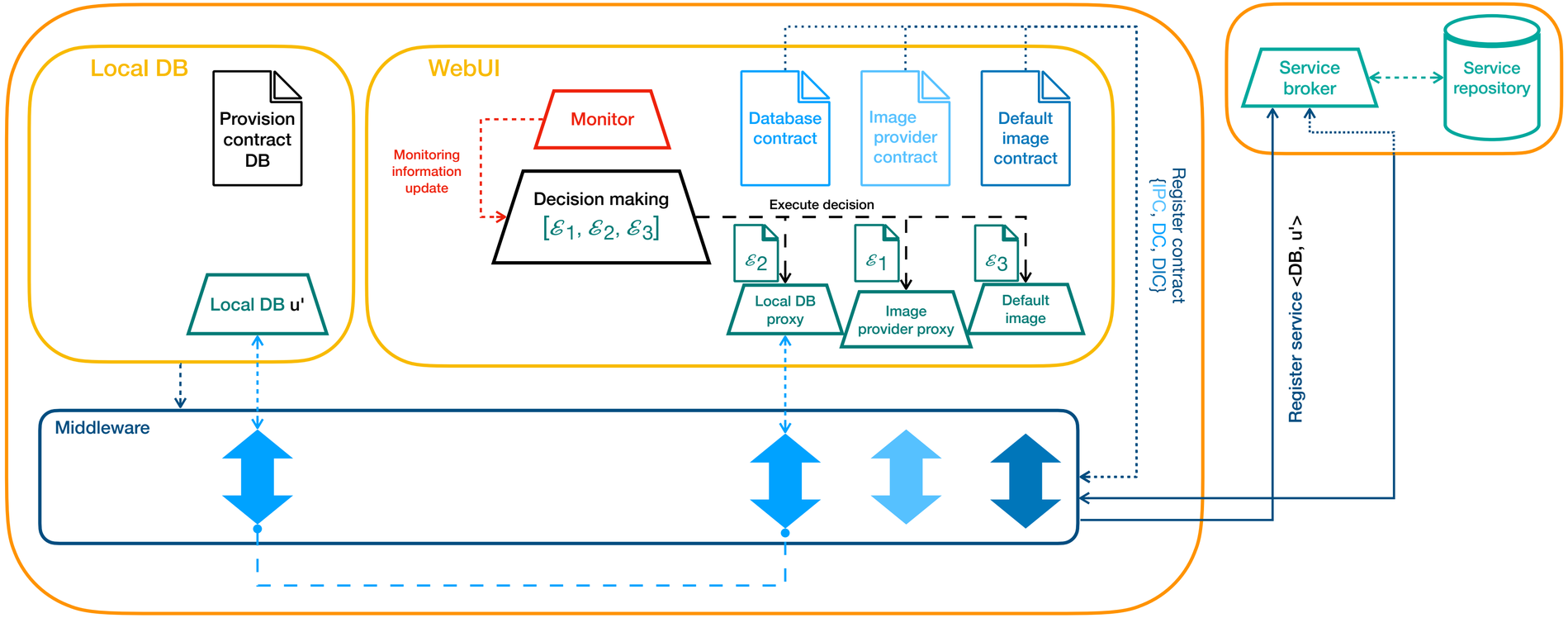

Local DB
Provision contract LocalDB
Local DB
WebUI
Monitor
Monitoring information update
Decision making
[$\mathcal{E}_1$, $\mathcal{E}_2$, $\mathcal{E}_3$]
Comm. contract LocalDB
Comm. contract ImgP
Comm. contract default
Execute decision
$\mathcal{E}_2$
$\mathcal{E}_1$
$\mathcal{E}_3$
Local DB
Image provider
Default image
Register contract
{ImgP, LocalDB, default}
Service broker
Service repository
Middleware
LocalDB
LocalDB
ImgP
default
Middleware
ImgP
Provision contract ImgP
Image provider

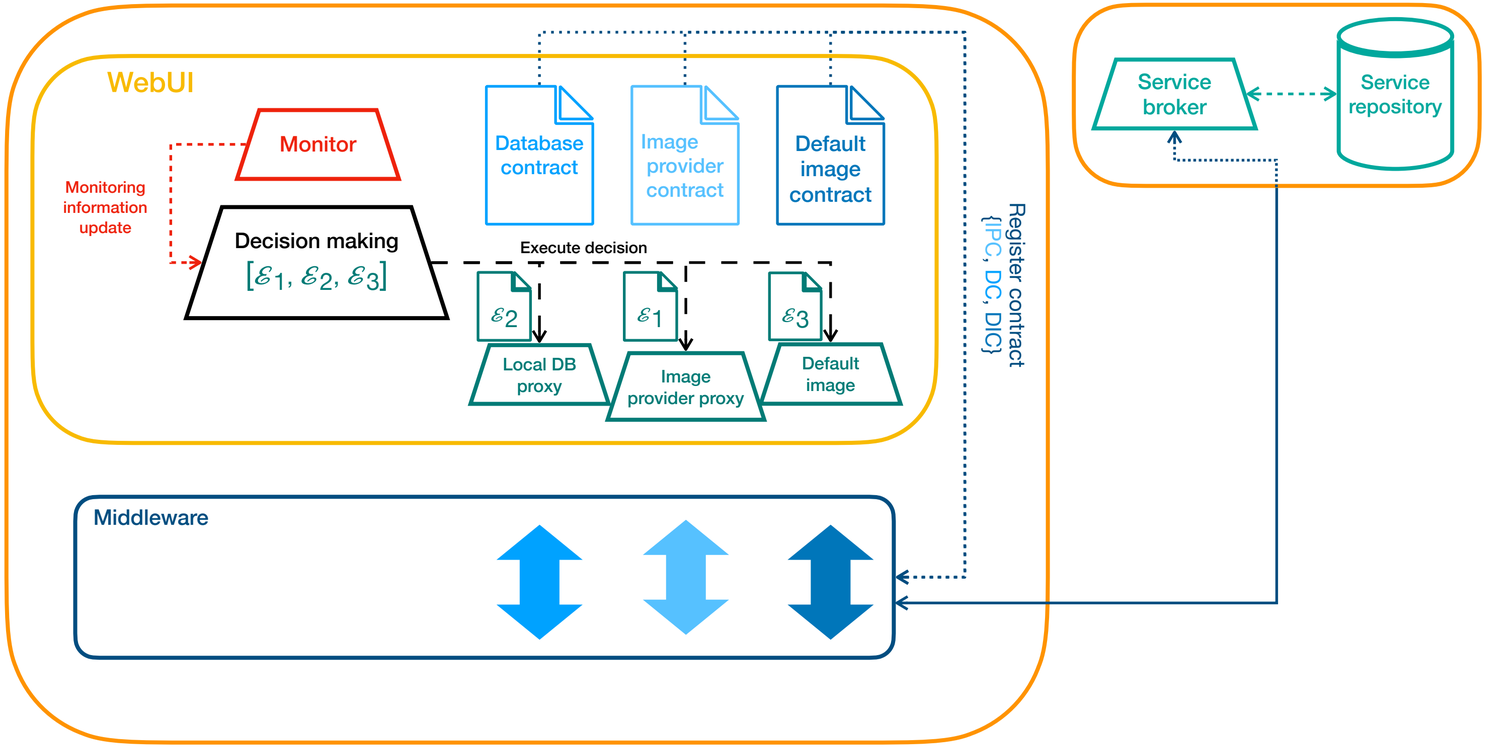

Local DB
Provision contract LocalDB
Local DB
WebUI
Comm. contract Monitor
Comm. contract LocalDB
Comm. contract ImgP
Comm. contract default
Decision making
[ℰ1, ℰ2, ℰ3]
Execute decision
ℰ2
ℰ1
ℰ3
Local DB
Image provider
Default image
Monitoring information update
Register contract
{Monitor, ImgP, LocalDB, default}
Middleware
LocalDB
Monitor
LocalDB
ImgP
default
Service broker
Service repository
Monitor
Middleware
Monitor
Middleware
ImgP
Provision contract ImgP
Image provider

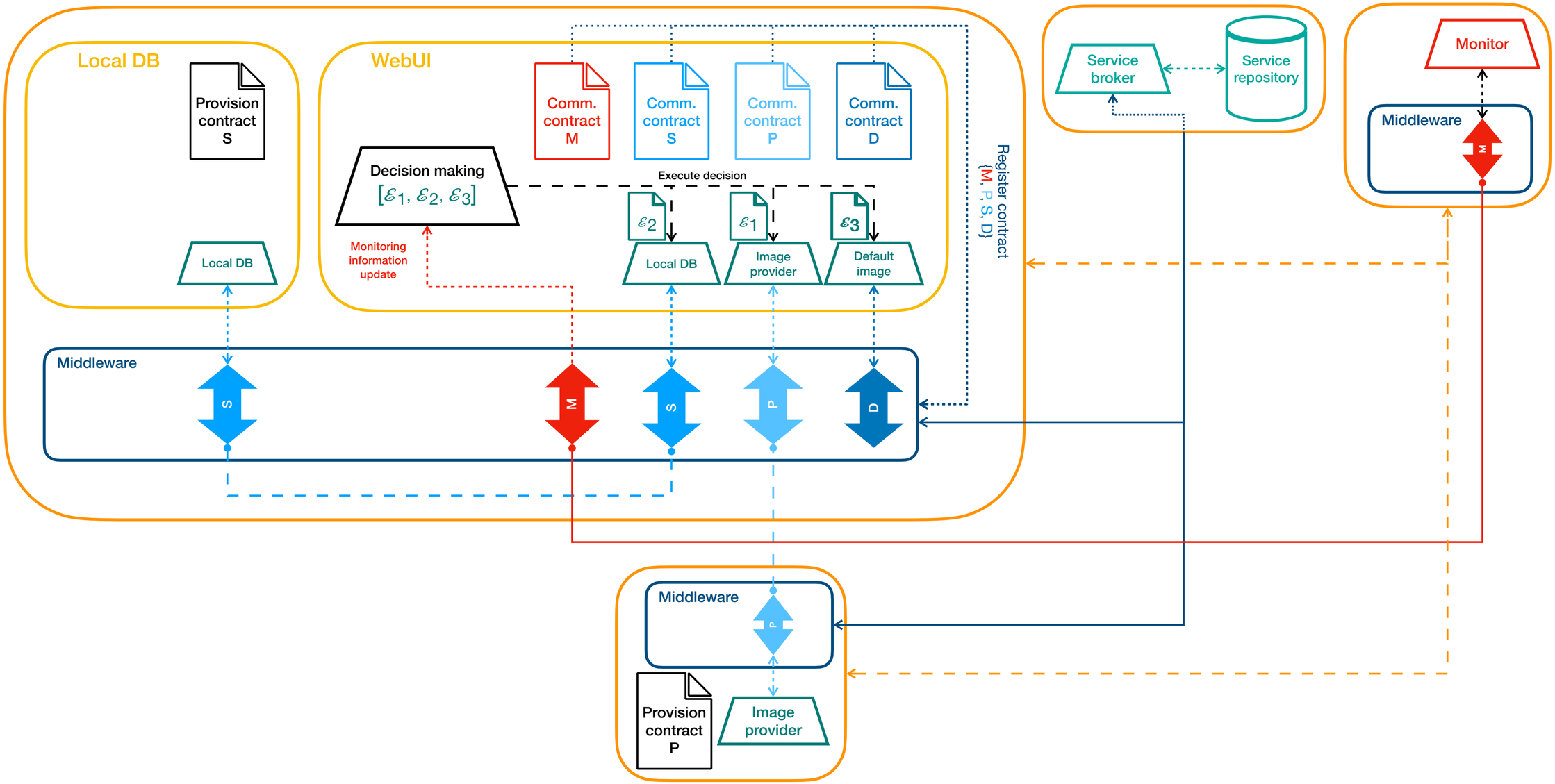

Local DB
Provision contract S
Local DB
WebUI
Monitor
Monitoring information update
Decision making
[ℰ1, ℰ2, ℰ3]
Comm. contract S
Comm. contract P
Comm. contract D
Execute decision
ℰ2
ℰ1
ℰ3
Local DB
Image provider
Default image
Register contract
{P, S, D}
Service broker
Service repository
Middleware
S
S
P
D
Middleware
P
Provision contract P
Image provider


Local DB
Provision contract DB
Local DB u'
WebUI
Monitor
Monitoring information update
Decision making
[ℰ1, ℰ2, ℰ3]
Database contract
Image provider contract
Default image contract
Execute decision
ℰ2
ℰ1
ℰ3
Local DB proxy
Image provider proxy
Default image
Register contract
{IPC, DC, DIC}
Service broker
Service repository
Register service <DB, u'>
Middleware
Middleware
Register service <ImgP, u>
Provision contract ImgP
Image provider u


WebUI
Monitor
Monitoring information update
Decision making
[$\mathcal{E}_1$, $\mathcal{E}_2$, $\mathcal{E}_3$]
Database contract
Image provider contract
Default image contract
Execute decision
$\mathcal{E}_2$
$\mathcal{E}_1$
$\mathcal{E}_3$
Local DB proxy
Image provider proxy
Default image
Register contract {IPC, DC, DIC}
Service broker
Service repository
Middleware
Middleware
Register service <ImgP, u>
Provision contract ImgP
Image provider u


Local DB
Provision contract DB
Local DB u'
WebUI
Monitor
Monitoring information update
Decision making
[ℰ1, ℰ2, ℰ3]
Database contract
Image provider contract
Default image contract
Execute decision
ℰ2
ℰ1
ℰ3
Local DB proxy
Image provider proxy
Default image
Middleware
Register contract
{IPC, DC, DIC}
Register service <DB, u'>
Service broker
Service repository


WebUI
Monitor
Monitoring information update
Decision making
[$\mathcal{E}_1$, $\mathcal{E}_2$, $\mathcal{E}_3$]
Database contract
Image provider contract
Default image contract
Execute decision
$\mathcal{E}_2$
$\mathcal{E}_1$
$\mathcal{E}_3$
Local DB proxy
Image provider proxy
Default image
Register contract
{IPC, DC, DIC}
Service broker
Service repository
Middleware


Local DB
Provision contract S
Local DB
WebUI
Comm. contract M
Comm. contract S
Comm. contract P
Comm. contract D
Decision making
[ℰ1, ℰ2, ℰ3]
Execute decision
ℰ2
ℰ1
ℰ3
Local DB
Image provider
Default image
Monitoring information update
Register contract {M, P, S, D}
Middleware
S
M
S
P
D
Service broker
Service repository
Monitor
Middleware
M
Middleware
P
Provision contract P
Image provider